

What we're doing

How good was the best contacts-v1 model we had on any given date — on contact R-precision and on held-out validation loss — and how tightly do those two track each other?

Every number exists somewhere already (#67, #75, #82, #89, #108, #117, #120, #146, #155, #160, #166, #169, #199/#204), but they are scattered across issue comments, per-experiment CSVs and two W&B projects, with three different inference recipes and two different validation-loss objectives mixed in. This experiment collects them into one dataset and three standing figures, and keeps them refreshable as new models land.

Why

Not a hypothesis-testing experiment — it is a **standing tracker**. The two things it is built to make visible, both of which prior issues assert piecemeal:

1. Progress on contact accuracy has come almost entirely from the **base model**, not from inference or post-training. 2. Validation loss predicts contact accuracy **across training generations** and stops predicting it **within a generation** (#169's finding), so the loss frontier and the accuracy frontier are not the same curve.

Results so far

The accuracy frontier moved in five jumps, all from the base model, none from inference or post-training: ~0.03 to 0.425 when #75's E8 rung finished (2026-06-21), 0.436 to 0.534 when #117's 16-epoch bs256 run finished (2026-07-22), 0.534 to 0.562 when #166's amino-acid augmentation continue-train of #117 finished (2026-07-31), 0.562 to 0.609 when #199's AFDB + ESM-Atlas sweep finished (2026-08-09/10), and 0.609 to 0.631 when that run's continuation was annealed to zero (2026-08-15).

The last two are both *data* changes rather than hyperparameter sweeps or epoch counts, and they are the two largest gains since #117. Between the first two jumps, five weeks of post-training and inference work moved the frontier by +0.011 (#120's re-epoch), and #160's backtracking fine-tune moved it -0.020.

Structure predictors on the same 554 proteins: Protenix-v2 single-seq 0.603, ESMFold 0.755, ESMFold2 0.786, Protenix-v2 + MSA 0.812.

The single-sequence gap has closed, and reversed

MarinFold's #199 CoreWeave cooldown scores 0.631 against Protenix-v2 single-sequence's 0.603 — a paired difference of +0.028 with a 95% CI of [+0.001, +0.054] over the 554 proteins.

The interval clears zero by 0.001, so read this as "ahead, barely" rather than a comfortable margin. But the sign has changed: two months ago the same comparison read 0.029 vs 0.603, and one model generation ago it was -0.016 with an interval that crossed zero.

MarinFold is higher on only 43% of individual proteins while leading on the mean — it is not winning more often, it is winning bigger where it wins, and those are the long proteins.

ESMFold2 (0.786) and MSA Protenix (0.812) are untouched by any of this, and the MSA gap has barely moved. That is where the remaining distance is.

#199 in context

#199 trains on AFDB plus 71.4B tokens of ESM-Atlas, and its best checkpoint is the first frontier point trained from scratch on CoreWeave H100s rather than continued from a #117 TPU checkpoint. Its from-scratch run is +0.047 over #166, and the cooldown that continues it adds another +0.022 for 29,040 further updates with the learning rate annealed to zero — no new data, no new hyperparameters. That model line was not finished when the sweep stopped it.

Two things it does not show. The CoreWeave/TRC gap (0.609 vs 0.524 at the same hyperparameter point) confounds initialization, schedule and step count, so it is not a platform result. And #199 has never been run against #155, the other ESM-Atlas result (0.554).

One number moved for a second reason: CW p06-aug reads 0.609 here, not the 0.587 published in #204. Same weights, same recipe — #234's harness and #208's independent check agree on ~0.609/0.610, and #234's reproduces the historical #75 checkpoint to 0.00017, so #204's reading of that one checkpoint is the outlier. It has not been explained.

#204 also re-evaluated the unchanged #117 checkpoint four times: they span 0.0023. That is this tracker's first noise estimate, and it makes #199's p03-base vs p03-aug difference (0.0036) a tie rather than a result.

Loss tracks accuracy across generations, and less each time

The exchange rate is collapsing. #75 E8 to #117 E16 bought 2.06 R-precision per nat; #117 to #166 bought 0.71; #166 to #199 CW bought about 0.33.

Inside one run it buys nothing at all: the 0.008-nat gap between #117's early-stop and final checkpoints is worth zero, and #146's 3B is 0.0012 better on loss and 0.023 worse on R-precision.

#204's sigmoid fit says the same thing from the other side — an upper asymptote of 0.5955, which #199 CW is already at 98.6% of. Either the relationship saturates near there, or the fit is extrapolating past its support.

So the loss frontier is useful for deciding which checkpoints are worth scoring, and useless for picking between two checkpoints of one run.

Two traps in these numbers

****Inference recipe.**** The same weights score ~ 0.086 higher under exp82's rollout recipe than under exp89's original pairwise scorer (#61/#75 E8: 0.339 to 0.425; #120: 0.350 to 0.436). The figures keep them visually separate. Eval-checkpoint skill output is pairwise before 2026-08-11 and rollout from that date on; anything from exp82's rollout workers or the exp169 dispatcher is rollout. Never infer the recipe from the magnitude. A third recipe, oracle best-of-100, is a diagnostic upper bound and never enters the frontier line.

****Loss scale.**** marin #7209 changed the packed-LM objective to mask padding targets, which raises the reported loss by ~ 0.38 nats. Everything through #166 is on the old scale; #199 is the first sweep on the new one. The figures plot the historical axis and convert #199 onto it, in green with a tilde. The conversion is good to about 0.025 nats — fine at the 0.1-nat scale, unreadable at the 0.01-nat scale. R-precision is unaffected.

Per-protein comparison with Protenix-v2

Over the 554 proteins, the #199 cooldown scores 0.631. Protenix-v2 single-sequence scores 0.603 (paired difference +0.028, MarFold higher on 43%). Protenix-v2 with MSAs scores 0.812 (paired difference -0.181, MarFold higher on 12%).

Re-pointed #117 to #166 to #199 CW to the cooldown, the shape did not change, only the gap: every length bin improved at each step and the single-sequence deficit went -0.069 to -0.041 to -0.016 to +0.028.

The two baselines trend opposite ways with length

Against single-sequence Protenix the gap narrows with length and now changes sign in three bins of four — everything above 100 residues: -0.076 below 100, +0.015 in 100-200, +0.074 in 200-400, +0.259 above 400.

Against MSA Protenix it widens: -0.165 below 100 residues, -0.333 above 400. Marifold does not win a single protein above 400 residues in that comparison.

The reason is in the marginals. Marifold declines with length (0.59 to 0.53) and single-sequence Protenix declines much faster (0.66 to 0.27), while MSA Protenix improves with length (0.75 to 0.86).

So "Marifold holds up better on long proteins" is about the single-sequence baseline only, it is a shallower decline rather than absolute strength, and the > 400 bin holds 17 proteins either way.

eval2, where the number is much smaller

The 554-protein set is ~75% designed proteins and not homology-controlled. On eval2 (#226) the same checkpoint scores 0.358 on the 78 natural proteins under 40% identity to anything in training, against Protenix-v2 single-sequence's 0.326 and ESMFold2's 0.529. Quote that cut when the question is generalization, and the 554 number only when the question is progress against everything published before it.

The unexplored lever

A diagnostic on #155's final checkpoint: scoring each of its 100 sampled rollouts per protein on its own best contacts, instead of voting them together, reads 0.595 — +0.041 of headroom that isn't reachable without ground truth, but points at reranking/selection as an unexplored lever.

That +0.041 is larger than the whole #199 CW to cooldown frontier step (+0.022), and the best model has meanwhile passed single-sequence Protenix, so selection is now headroom beyond the baseline rather than a way to reach it. The comparison crosses two checkpoints, so treat it as a rough ordering rather than arithmetic.

Open gap

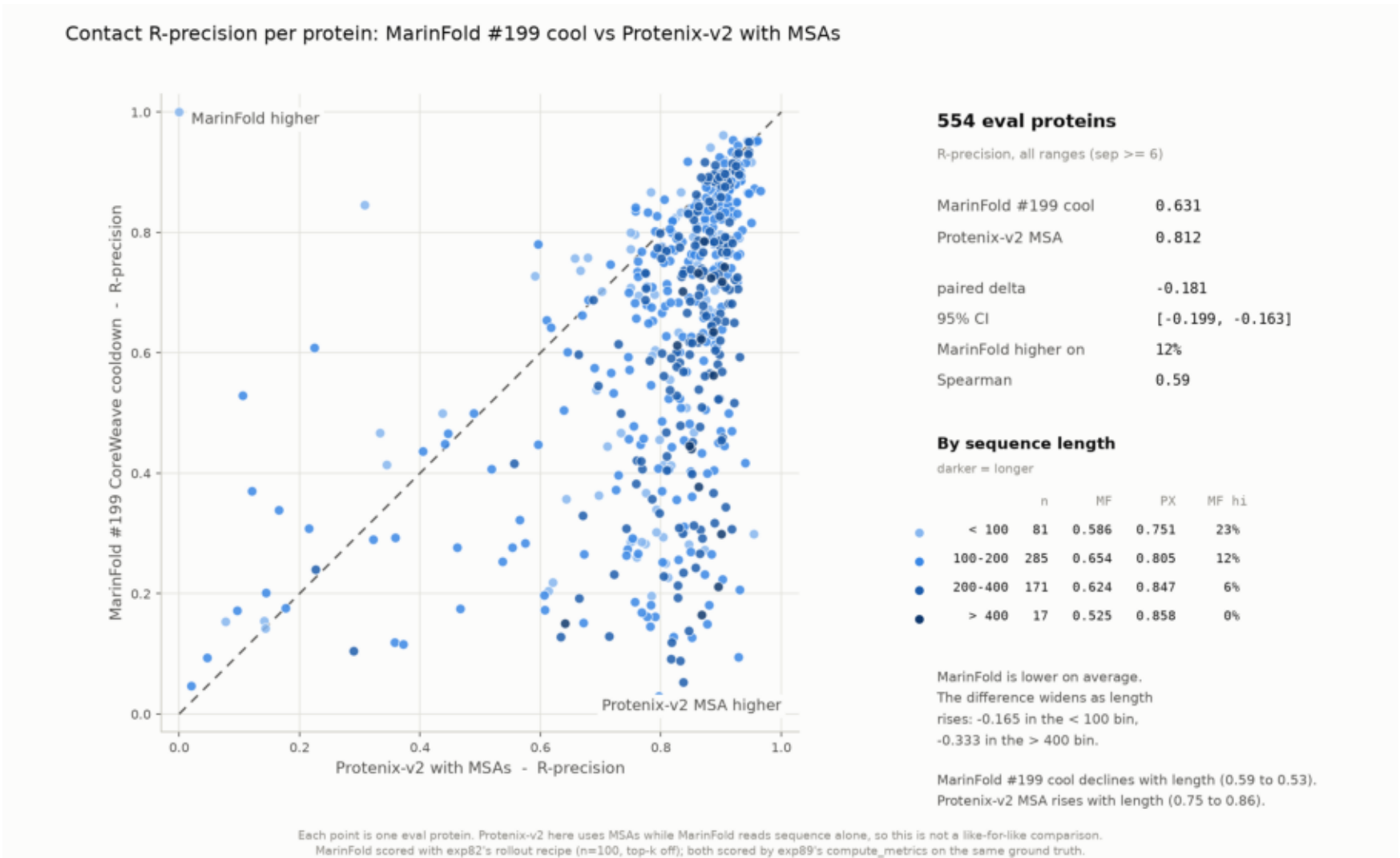
#108's 3B on CoreWeave held the loss frontier from 2026-07-11 to 07-16 at 2.7418 and was never contact-scored. Given #146 — a 3B at matched loss under-performing the 1.5B — it is probably not a missed frontier point, but that is an inference, not a measurement.

Keeping this current

Two commands: `build_dataset.py` re-pulls W&B, `plot_progress.py` redraws (plus `plot_vs_protenix.py` for the head-to-head pair). New benchmark scores are hand-added to `RPRECISION_ROWS` with a source citation, an explicit inference recipe and an explicit loss scale; a new W&B sweep tag goes in `WANDB_SOURCES` with its scale — read that off the run's own `requirements.txt`, not its date. When a new model takes the frontier, re-point `MARINFOLD_MODEL` and `ROWS_CSV` in `plot_vs_protenix.py` — it needs published per-protein rows, not a summary. Full procedure in the README.

marifold_vs_protenix_msa.png

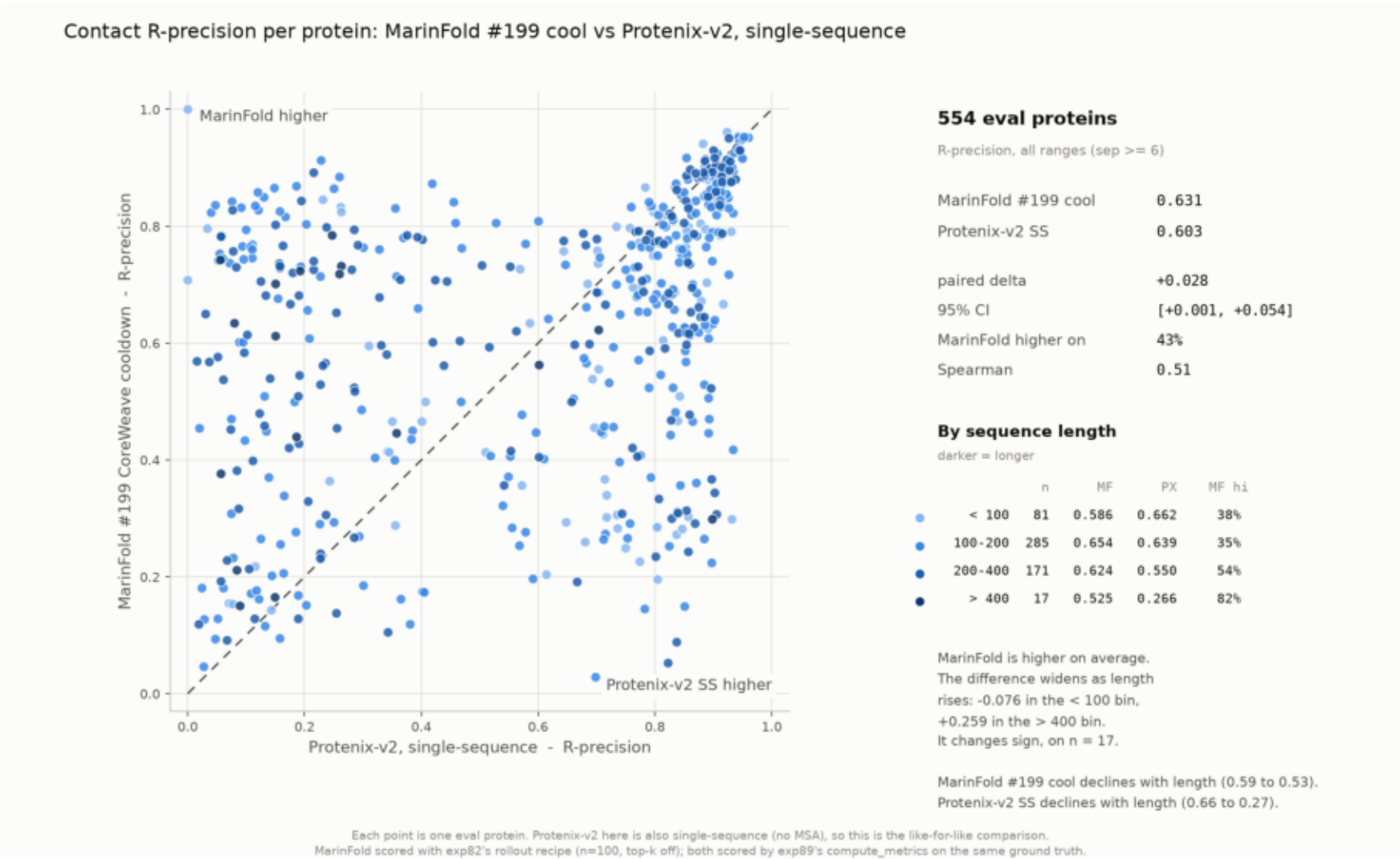
Per-protein R-precision, MarInFold #199 cool vs Protenix-v2 MSA (n=554). Means 0.631 vs 0.812; the difference varies with length.



```
$ plot_vs_protenix.py --baseline msa
```

marinfold_vs_protenix_ss.png

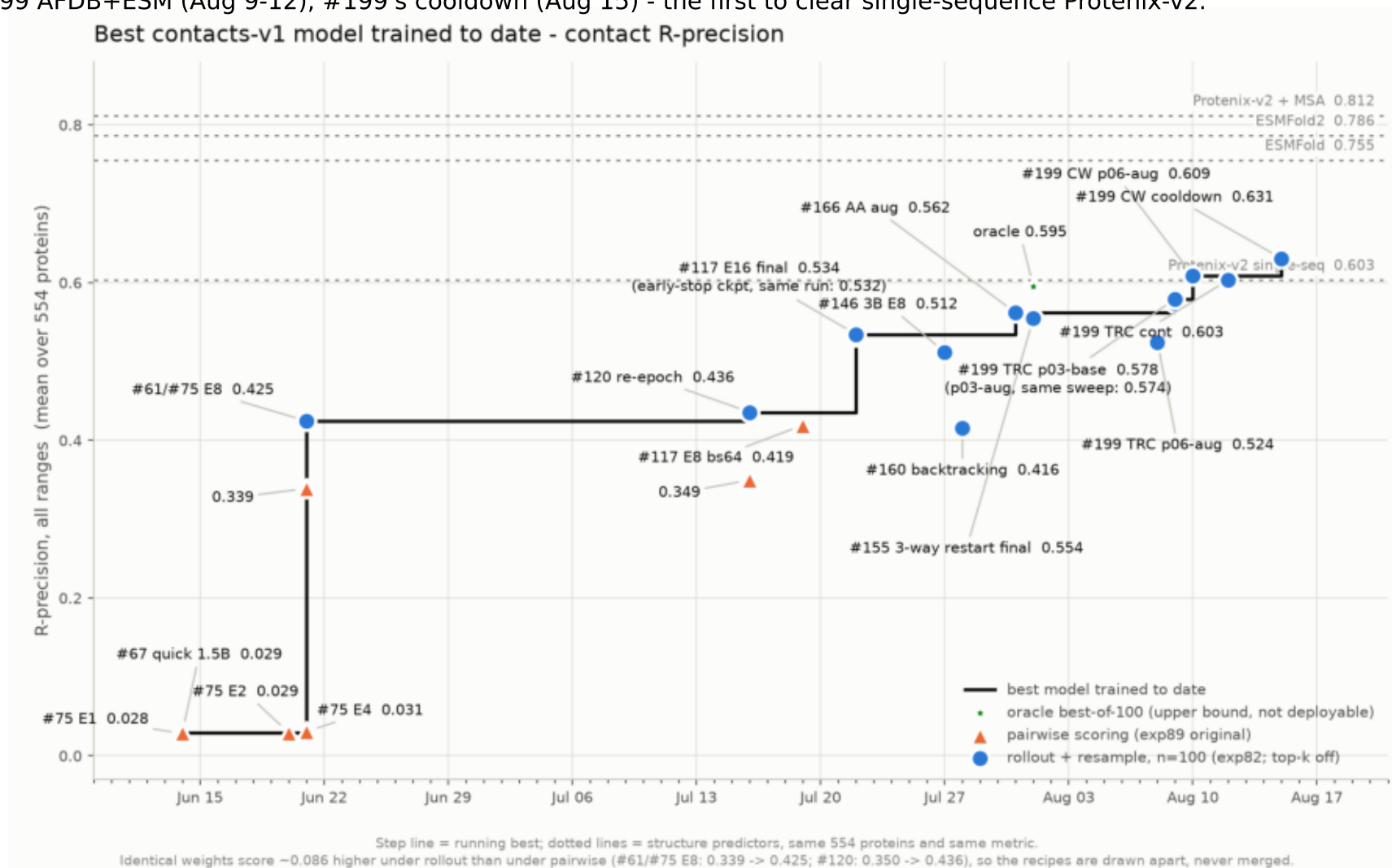
Per-protein R-precision, MarinFold #199 cool vs Protenix-v2 SS (n=554). Means 0.631 vs 0.603; the difference varies with length.



```
$ plot_vs_protenix.py --baseline single_seq
```

rprecision_frontier.png

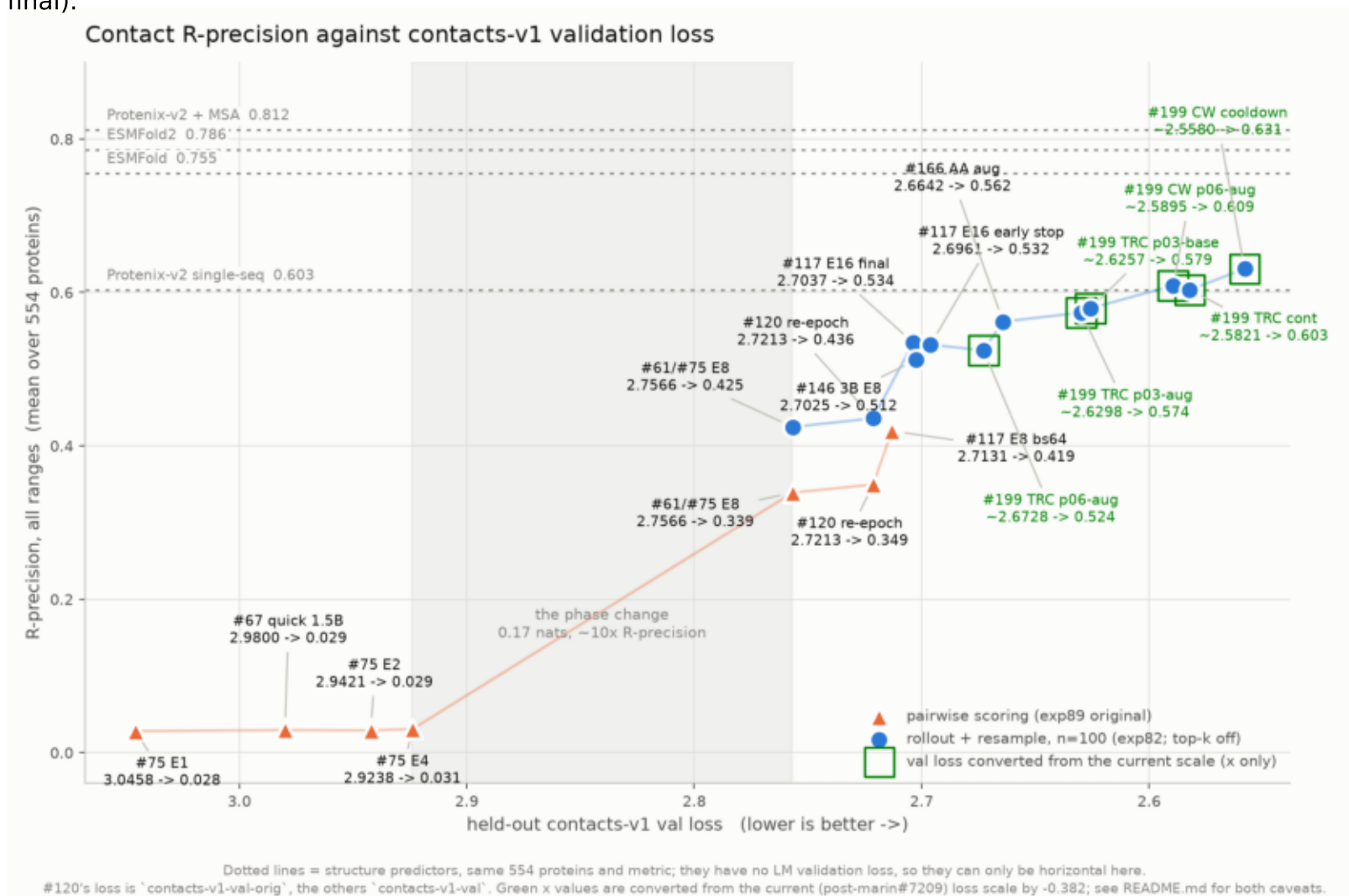
Best contacts-v1 model to date on contact R-precision. Every step is a base-model result: #75 E8 (Jun 21), #117 E16 (Jul 22), #166 AA aug (Jul 31), #199 AFDB+ESM (Aug 9-12), #199's cooldown (Aug 15) - the first to clear single-sequence Protenix-v2.



\$ plot_progress.py

rprecision_vs_val_loss.png

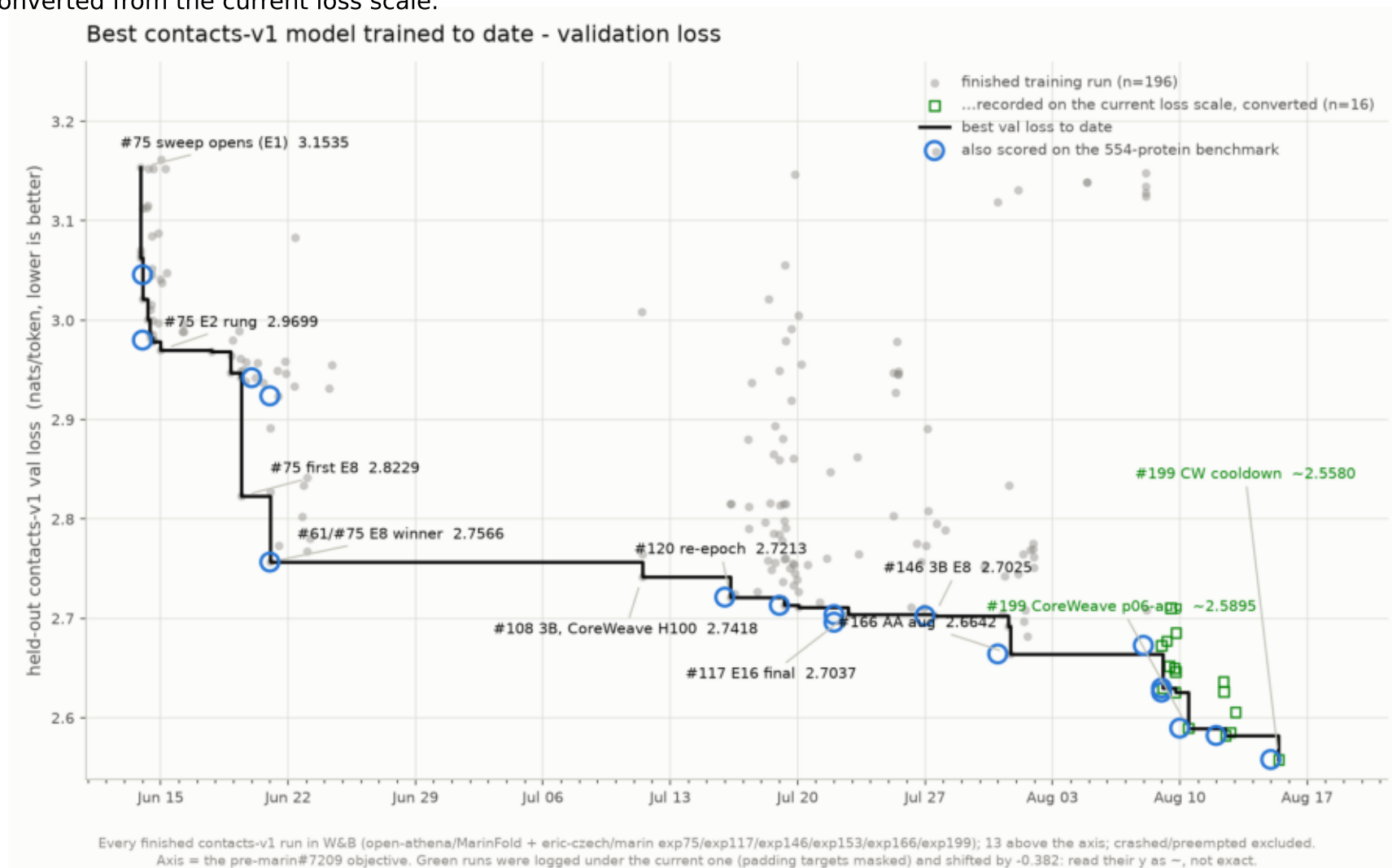
R-precision against val loss. Across generations the exchange rate is collapsing (2.06 -> 0.71 -> ~0.33 R per nat); inside one run it is flat (#117 early vs final).



\$ plot_progress.py

val_loss_frontier.png

Best held-out contacts-v1 val loss to date over 196 finished runs. The loss frontier has many more steps than the accuracy one; the last three are converted from the current loss scale.



\$ plot_progress.py